

MS 6015: Multivariate Statistical Methods

Sampling Distributions and Inference about the Mean Vector

Condensed Lecture Notes — Reference: Johnson & Wichern, Chapters 4–6, Keller
Chapter 14

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1 Properties of the Multivariate Normal Distribution

1.1 Density Function

A p -dimensional random vector $\mathbf{X}$ follows a multivariate normal distribution, written $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with density:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right\}, \quad x_i \in \mathbb{R}.$$

1.2 Key Properties

Properties of $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

1. Linear combinations of the components of $\mathbf{X}$ are (multivariate) normal.
2. Every subset of the components of $\mathbf{X}$ is (multivariate) normal.
3. Zero covariance between components implies statistical independence.
4. Conditional distributions of the components are (multivariate) normal.

1.3 Conditional Distribution

Partition $(\mathbf{X}_1^{(q_1 \times 1)}, \mathbf{X}_2^{(q_2 \times 1)}) \sim N_{q_1+q_2}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

Then the conditional distribution of $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2$ is multivariate normal with:

$$\text{Mean: } \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \quad \text{Covariance: } \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}.$$

2 Sampling Distributions of $\bar{X}$ and S

Sampling Distribution Results

- **Sample mean vector:** $\bar{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}/n)$.
- **Sample covariance:** $(n-1)\mathbf{S}$ follows a *Wishart distribution* with $n-1$ degrees of freedom (the multivariate analogue of χ^2).
- $\bar{X}$ and $\mathbf{S}$ are **independent**.

2.1 Central Limit Theorem (Multivariate)

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be i.i.d. from *any* population with mean $\boldsymbol{\mu}$ and finite covariance $\boldsymbol{\Sigma}$. Then:

$$\sqrt{n}(\bar{X} - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

3 Univariate Hypothesis Testing (Review)

3.1 One-Sample t -Test

Test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0.$$

Reject H_0 at level α when $|t| > t_{n-1}(\alpha/2)$. Equivalently, reject when

$$t^2 = n(\bar{x} - \mu_0)(s^2)^{-1}(\bar{x} - \mu_0) > t_{n-1}^2(\alpha/2).$$

A $100(1 - \alpha)\%$ confidence interval for μ is:

$$\bar{x} \pm t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}}.$$

3.2 Common Univariate Tests (Summary)

Test	Statistic	Distribution
One-sample t (unknown σ)	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	t_{n-1}
One-sample z (known σ)	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	$N(0, 1)$
Two-sample F (variance ratio)	$F = s_1^2/s_2^2$	F_{n_1-1, n_2-1}
One-Way ANOVA (k means)	$F = \text{MST}/\text{MSE}$	$F_{k-1, n-k}$

4 Hotelling's T^2 Statistic

4.1 Definition

The multivariate generalisation of the squared t -statistic is:

$$T^2 = n(\bar{X} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{X} - \boldsymbol{\mu}_0).$$

This is the **Mahalanobis distance** between $\bar{X}$ and $\boldsymbol{\mu}_0$:

- $(\bar{X} - \boldsymbol{\mu}_0)$ is the mean-difference vector (signal).
- $\mathbf{S}^{-1}$ downweights directions with high variability.

4.2 Sampling Distribution and Decision Rule

Under $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ with multivariate normality:

$$\frac{n-p}{p(n-1)} T^2 \sim F_{p, n-p}.$$

Rejection Rule

Reject H_0 at level α when:

$$T^2 > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha).$$

Requires $n > p$ so that $\mathbf{S}$ is invertible. When $p = 1$, this reduces to the usual t -test.

4.3 Invariance Property

T^2 is invariant under linear transformations $\mathbf{Y} = C\mathbf{X} + \mathbf{d}$ (with C non-singular), so the test result does not depend on the units of measurement.

Female Sweat Data ($n = 20, p = 3$)

Testing $H_0 : \boldsymbol{\mu}_0 = [4, 50, 10]^\top$ for sweat rate, sodium, and potassium.

At $\alpha = 0.10$: critical value $= \frac{19.3}{17} F_{3,17}(0.10) = 8.18$.

Computed $T^2 = 9.74 > 8.18 \Rightarrow$ **Reject** H_0 .

Women's Nutritional Survey ($n = 737, p = 5$)

Testing whether average dietary intake meets federal guidelines.

$T^2 = 1758.54$, F -statistic $= 349.80 > 3.042 = F_{5,732}(0.01) \Rightarrow$ **Reject** H_0 .

Conclusion: Women fall short on calcium and iron, and may exceed recommendations for protein, Vitamin A, and Vitamin C.

5 Confidence Regions and Simultaneous Confidence Intervals

5.1 Confidence Ellipsoid

A $100(1 - \alpha)\%$ confidence region for $\boldsymbol{\mu}$ is the ellipsoid:

$$\left\{ \boldsymbol{\mu} : n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \leq \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right\}.$$

This is a p -dimensional ellipsoid centred at $\bar{\mathbf{x}}$, with axes oriented along the eigenvectors of $\mathbf{S}$ and half-lengths proportional to $\sqrt{\lambda_i/n}$, where λ_i are eigenvalues of $\mathbf{S}$.

5.2 Three Types of Confidence Intervals

Comparison of CI Methods for μ_j

1. One-at-a-Time (Marginal):

$$\bar{x}_j \pm t_{n-1, \alpha/2} \sqrt{s_{jj}/n}.$$

Each individual CI has coverage $1 - \alpha$, but the *joint* coverage is less than $1 - \alpha$.

2. Bonferroni Simultaneous CIs:

$$\bar{x}_j \pm t_{n-1, \alpha/(2p)} \sqrt{s_{jj}/n}, \quad j = 1, \dots, p.$$

Joint coverage $\geq 1 - \alpha$. Conservative (wider intervals), but simple and assumption-free.

3. Hotelling T^2 Simultaneous CIs:

$$\bar{x}_j \pm \sqrt{\frac{p(n-1)}{n-p} F_{p, n-p}(\alpha)} \cdot \sqrt{s_{jj}/n}, \quad j = 1, \dots, p.$$

Derived from the confidence ellipsoid. Exact joint coverage $1 - \alpha$ under normality; often less conservative than Bonferroni.

Rule of thumb: Use Bonferroni when you only care about a few specific components; use T^2 intervals when you want inference for all components simultaneously.

6 Likelihood Ratio Test and Wilks' Lambda

The Likelihood Ratio statistic compares the restricted likelihood (at μ_0) to the unrestricted MLE:

$$\Lambda = \left(\frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|} \right)^{n/2}.$$

Wilks' Lambda is defined as:

$$\Lambda^{2/n} = \frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|} = \left(1 + \frac{T^2}{n-1} \right)^{-1}.$$

Small values of $\Lambda^{2/n}$ (equivalently large T^2) lead to rejection of H_0 .

7 Two-Sample Tests

7.1 Univariate: Two Independent Samples

To test $H_0 : \mu_1 = \mu_2$ (or $H_0 : \mu_1 - \mu_2 = \delta_0$):

Case 1 — Equal variances ($\sigma_1^2 = \sigma_2^2$):

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{s_p \sqrt{1/n_1 + 1/n_2}} \sim t_{n_1 + n_2 - 2}.$$

Case 2 — Unequal variances ($\sigma_1^2 \neq \sigma_2^2$), Welch:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}, \quad \nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

7.2 Multivariate: Two Independent Samples

Assumptions: Two independent random samples from $N_p(\boldsymbol{\mu}_l, \boldsymbol{\Sigma})$, $l = 1, 2$, with $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}$.

Pooled covariance estimator:

$$\mathbf{S}_{\text{pool}} = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}.$$

Two-sample T^2 :

$$T^2 = [(\bar{X}_1 - \bar{X}_2) - \boldsymbol{\delta}_0]^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{\text{pool}} \right]^{-1} [(\bar{X}_1 - \bar{X}_2) - \boldsymbol{\delta}_0].$$

Distributed as $\frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}$ under H_0 .

Large samples ($n_1 - p$ and $n_2 - p$ large): No need to assume $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$ or normality; $T^2 \xrightarrow{d} \chi_p^2$.

8 Paired Comparisons

8.1 When to Use Paired Design

Use a paired comparison when there is a natural one-to-one correspondence between observations in the two groups (e.g., pre/post measurements, matched subjects). Pairing reduces extraneous variability and typically yields greater power than independent samples.

8.2 Univariate Paired t -Test

Compute differences $D_j = X_{j1} - X_{j2}$ and test $H_0 : \delta = 0$:

$$t = \frac{\bar{D}}{s_D / \sqrt{n}} \sim t_{n-1} \quad \text{under } H_0.$$

8.3 Multivariate Paired Comparison

Form difference vectors $\mathbf{D}_j = \mathbf{X}_{j1} - \mathbf{X}_{j2}$. Assuming $\mathbf{D}_j \sim N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_D)$ i.i.d.:

$$T^2 = n \bar{\mathbf{D}}^\top \mathbf{S}_D^{-1} \bar{\mathbf{D}} \sim \frac{(n-1)p}{n-p} F_{p, n-p} \quad \text{under } H_0 : \boldsymbol{\delta} = \mathbf{0}.$$

Reject H_0 if $T^2 \geq \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$.

9 One-Way Analysis of Variance (ANOVA)

9.1 Setup

k independent samples (treatments) are drawn. Let x_{lj} be the j -th observation in treatment l , and $\bar{x}$ be the grand mean.

Hypotheses: $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ vs. H_1 : at least two means differ.

9.2 Decomposition and Test Statistic

$$\underbrace{\sum_{l,j} (x_{lj} - \bar{x})^2}_{\text{SS(Total)}} = \underbrace{\sum_l n_l (\bar{x}_l - \bar{x})^2}_{\text{SST (Between)}} + \underbrace{\sum_{l,j} (x_{lj} - \bar{x}_l)^2}_{\text{SSE (Within)}}.$$

$$F = \frac{\text{MST}}{\text{MSE}} = \frac{\text{SST}/(k-1)}{\text{SSE}/(n-k)} \sim F_{k-1, n-k} \quad \text{under } H_0.$$

ANOVA Table

Source	SS	df	MS	F
Treatments	SST	$k-1$	MST	MST/MSE
Error	SSE	$n-k$	MSE	
Total	SS(Total)	$n-1$		

9.3 Why Not Pairwise t -Tests?

For $k = 6$ groups there are $\binom{6}{2} = 15$ pairs. If each is tested at $\alpha = 0.05$, the probability of at least one Type I error is $1 - (0.95)^{15} \approx 0.54$. ANOVA controls the global error rate.

9.4 Post-Hoc Multiple Comparisons

When H_0 is rejected, determine *which* pairs differ:

- **Fisher's LSD:** Least significant difference; most powerful but no family-wise control.
- **Bonferroni:** Divide α by the number of comparisons.
- **Tukey's method:** Controls family-wise error rate exactly for all pairwise comparisons.

9.5 Required Conditions

- Normality within each group (check via histograms).
- Equal population variances (check by comparing sample variances).
- Non-parametric alternative: **Kruskal-Wallis test**.

10 Two-Way ANOVA: Randomized Block Design (RBD)

The RBD extends ANOVA by grouping experimental units into b homogeneous blocks (analogous to matched-pairs for $k > 2$ treatments).

$$\text{SS(Total)} = \underbrace{\text{SST}}_{\text{Between treatments}} + \underbrace{\text{SSB}}_{\text{Between blocks}} + \underbrace{\text{SSE}}_{\text{Residual}}.$$

Blocking removes between-block variability from SSE, making it easier to detect treatment differences. You can also formally test whether blocks differ ($F = \text{MSB}/\text{MSE}$).

11 Two-Factor ANOVA (Factorial Design)

Two factors A (a levels) and B (b levels), with n balanced replicates per cell.

Model: $X_{lkr} = \mu + \tau_l + \beta_k + \gamma_{lk} + \epsilon_{lkr}$, where γ_{lk} is the AB interaction.

SST is partitioned:

$$SST = SS(A) + SS(B) + SS(AB).$$

Three Hypotheses in Two-Factor ANOVA

- **Factor A:** H_0 : means at all a levels of A are equal; $F = MS(A)/MSE$.
- **Factor B:** H_0 : means at all b levels of B are equal; $F = MS(B)/MSE$.
- **Interaction AB:** H_0 : A and B do not interact; $F = MS(AB)/MSE$.

Interaction means the effect of factor A depends on the level of factor B.

12 One-Way MANOVA

12.1 Motivation

When multiple correlated outcome variables are measured simultaneously, analysing them jointly:

- Accounts for correlations between variables.
- Controls the family-wise Type I error rate across variables.
- Yields greater power than running separate ANOVAs.

12.2 MANOVA Model

$$\mathbf{X}_{lj} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\epsilon}_{lj}, \quad \boldsymbol{\epsilon}_{lj} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}) \text{ i.i.d.}, \quad \sum_{l=1}^k n_l \boldsymbol{\tau}_l = \mathbf{0}.$$

12.3 SSCP Decomposition

$$\underbrace{\mathbf{T}}_{\text{Total SSCP}} = \underbrace{\mathbf{H}}_{\text{Between (Treatment)}} + \underbrace{\mathbf{W}}_{\text{Within (Error)}},$$

where

$$\mathbf{H} = \sum_{l=1}^k n_l (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})(\bar{\mathbf{x}}_l - \bar{\mathbf{x}})^\top, \quad \mathbf{W} = \sum_{l=1}^k (n_l - 1) \mathbf{S}_l.$$

12.4 Test Statistics

MANOVA Test Statistics

1. **Wilks' Lambda** (most common):

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{T}|} = \frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{H}|}.$$

Small $\Lambda^* \Rightarrow$ reject H_0 .

2. **Hotelling–Lawley Trace:** $\text{tr}(\mathbf{W}^{-1}\mathbf{H})$. Reject for large values.
3. **Pillai's Trace:** $\text{tr}[\mathbf{H}(\mathbf{H} + \mathbf{W})^{-1}]$. Most robust to non-normality.
4. **Roy's Maximum Root:** Largest eigenvalue of $\mathbf{W}^{-1}\mathbf{H}$. Best power when only one eigenvalue is non-zero.

Rule: If all four statistics agree, report Wilks' Λ^* . If they disagree, investigate why (usually due to the structure of group differences).

12.5 Relationship to Wilks' Lambda and the F-Distribution

For special cases, Λ^* has exact F -distributions:

p	k (groups)	Transformed statistic $\sim F$
1	any	$F_{k-1, n-k}$
2	any	$F_{2(k-1), 2(n-k-1)}$
any	2	$F_{p, n-p-1}$
any	3	$F_{2p, 2(n-p-2)}$

13 Two-Way MANOVA (Randomized Block Design)

Extends the RBD to multivariate responses. The additive model is:

$$\mathbf{X}_{lj} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\beta}_j + \boldsymbol{\epsilon}_{lj},$$

with the assumption of no block-by-treatment interaction.

$$\mathbf{T} = \mathbf{H} + \mathbf{B} + \mathbf{W}.$$

MANOVA Table (RBD)

Source	df	SSCP	Test
Blocks	$b - 1$	$\mathbf{B}$	$\Lambda^* = \mathbf{W} / \mathbf{W} + \mathbf{B} $
Treatment	$k - 1$	$\mathbf{H}$	$\Lambda^* = \mathbf{W} / \mathbf{W} + \mathbf{H} $
Error	$(b - 1)(k - 1)$	$\mathbf{W}$	
Total	$bk - 1$	$\mathbf{T}$	

14 Two-Factor MANOVA

Multivariate extension of two-factor ANOVA with a levels of Factor A, b levels of Factor B, and n replicates per cell.

Model with interaction:

$$\mathbf{X}_{lkr} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{lk} + \boldsymbol{\epsilon}_{lkr}.$$

Three Hypotheses in Two-Factor MANOVA

- **Factor A:** $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_a = \mathbf{0}$. Test with $\Lambda^* = |\mathbf{W}|/|\mathbf{W} + \mathbf{H}_A|$.
- **Factor B:** $H_0 : \boldsymbol{\beta}_1 = \cdots = \boldsymbol{\beta}_b = \mathbf{0}$. Test with $\Lambda^* = |\mathbf{W}|/|\mathbf{W} + \mathbf{H}_B|$.
- **Interaction:** $H_0 : \boldsymbol{\gamma}_{lk} = \mathbf{0}$ for all l, k . Test with $\Lambda^* = |\mathbf{W}|/|\mathbf{W} + \mathbf{H}_{AB}|$.

15 Grand Summary Table

Scenario	Univariate	Multivariate
One sample vs. known target	One-sample t -test $H_0 : \mu = \mu_0$	Hotelling's T^2 $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$
Two independent samples	Two-sample t -test $H_0 : \mu_1 = \mu_2$	Two-sample T^2 $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$
Paired samples	Paired t -test	Paired T^2 on differences
k groups, 1 factor	One-Way ANOVA	One-Way MANOVA
k treatments + blocks	RBD ANOVA	Two-Way MANOVA (RBD)
Two factors + interaction	Two-Factor ANOVA	Two-Factor MANOVA

Key Conceptual Parallels

- t -statistic $\longleftrightarrow T^2$ (Mahalanobis distance).
- χ^2 distribution $\longleftrightarrow$ Wishart distribution.
- F -ratio (ANOVA) $\longleftrightarrow$ Wilks' Λ^* (MANOVA).
- Confidence interval $\longleftrightarrow$ Confidence ellipsoid.
- Sample variance $s^2 \longleftrightarrow$ Sample covariance matrix $\mathbf{S}$.